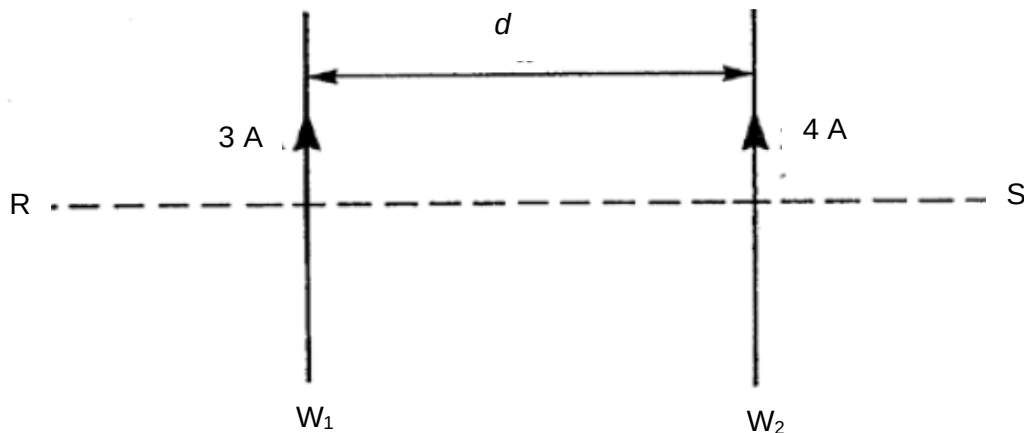


- 22 Two vertical wires W_1 and W_2 are distance d apart carrying currents of 3 A and 4 A respectively.



The combined magnetic fields due to these currents gives a position of zero magnetic flux density on the horizontal line RS passing through the wires.

What is the distance of this position from the wire W_1 ?

A $\frac{1}{4}d$

B $\frac{3}{7}d$

C $\frac{4}{7}d$

D $\frac{3}{4}d$

Ans: B

For zero magnetic flux density, $B_1 = B_2$

$$B_1 = \mu_0 I_1 / 2\pi x$$

$$B_2 = \mu_0 I_2 / 2\pi(d - x)$$

$$\mu_0 I_1 / 2\pi x = \mu_0 I_2 / 2\pi(d - x) \rightarrow (3)/x = (4)/(d - x) \rightarrow 3d - 3x = 4x \rightarrow x = 3/7 d$$

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A uniform magnetic field of flux density B passes normally through a plane area A . In this